much loved APJ Abdul Kalam. Till date, the new organism – a form of bacteria – has been found only on the International Space Station (ISS) and has been found on earth. Researchers at the Jet Propulsion Laboratory (JPL) the foremost lab of NASA for work on inter-planetary travel discovered the new bacteria on the filters of the international space station (ISS) and named it Solibacillus kalam to honour the late president, who was a renowned aerospace scientist.

HOTS

What could be source of microorganisms in space?

1.2 Emerging Microbes

Microbes and the diseases they cause have emerged as a major threat to human populations. Some of them like malaria have been major pathogens since they emerged in the depths of antiquity.

Emerging infection is defined as those infections whose incidence in humans has increased in the past two decades or will increase in the future. It can be new, reemerging, drug resistant infections.

Ebola

Ebola virus disease (EVD), also known as **Ebola hemorrhagic fever (EHF)** or simply **Ebola**, is a viral hemorrhagic fever of humans and other primates caused by Ebola virus.

Ebola is a rare but deadly virus that causes fever, body aches, and diarrhea, and sometimes bleeding inside and outside the body. The virus spreads through the body; it damages the immune system and organs. Ultimately, it causes levels of blood-clotting cells to drop. This leads to severe, uncontrollable bleeding. The disease was known as Ebola hemorrhagic fever but is now referred to as Ebola virus. It kills up to 90% of people who are infected.

Ebola is a contagious as more common viruses like colds, influenza, or measles. It spreads to people by contact with the skin or bodily fluids of an infected animal, like a monkey, chimp, or fruit bat. Then it moves from person to person the same way early on, Ebola can feel like the flu or other illnesses. Symptoms show up 2 to 21 days after infection and usually include: High fever, Headache, Joint and muscle aches, Sore throat, Weakness, Stomach pain, Lack of appetite. It causes bleeding inside the body, as well as from the eyes, ears, and nose. Some people will vomit or cough up blood, have bloody diarrhea, and get a rash (Figure 1.1).



Figure 1.1: Ebola virus

Zika

Zika virus (ZIKV) is a member of the virus family *Flaviviridae*. It is spread by daytime-active *Aedes* mosquitoes, such as *A. aegypti* and *A. albopictus*. Its name comes from the Zika Forest of Uganda, where the virus was first

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